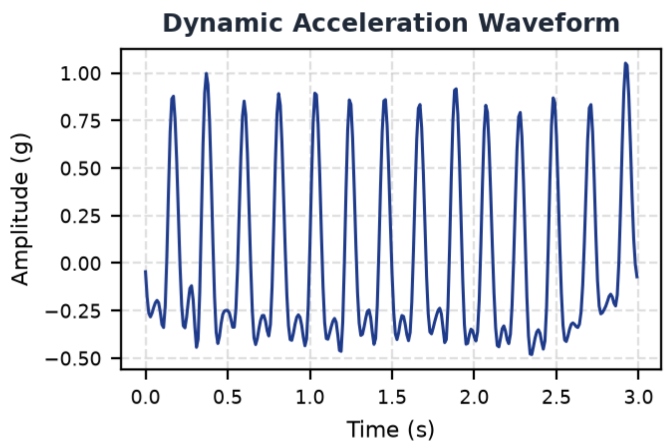
NEUROTRACK — Tremor Assessment & Telemetry Report

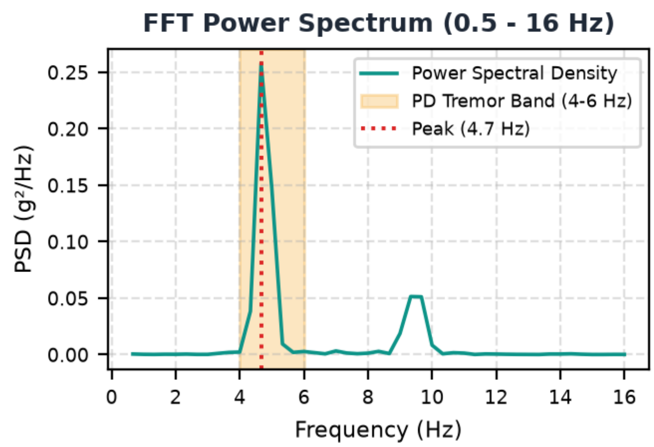
Report ID: NT-1788584487
Date: 2026-09-05 10:31:27

Patient Identifier:	LIVE_COM4	**Data Origin:**	**■ Live Physical Hardware (ESP32/COM4)**
Sensor Device:	Physical ESP32 + MPU6050 (COM4)	**Sampling Rate:**	50.0 Hz (Nyquist: 25.0 Hz)
Assessment Duration:	6.0 seconds	**Analysis Window:**	3.0 sec (50% sliding overlap)

AI Pattern Classification	Tremor Severity Index	Clinical Severity Grade
PD Confidence: 99.2% (PD Match: 99.2%)	**99.7** / 100	**Marked / Severe** High-amplitude sustained resting tremor with strong rotational and kinematic harmonic components.

Biomechanical Time-Series & Spectral Isolation





Extracted Digital Biomarkers (Window Metrics)

Biomarker	Observed Value	Reference Band / Diagnostic Context
Dominant Frequency	**4.67 Hz**	Parkinsonian Resting Tremor: 4.0 - 6.0 Hz
Tremor Band Power Ratio	**75.1%**	Ratio of power in 4-6 Hz band to broad spectrum (>40% indicates localized tremor)
Dynamic Jerk RMS	**17.31 g/s**	Time-derivative of acceleration measuring abrupt oscillatory changes
Spectral Shannon Entropy	**0.482**	Low (<0.45) indicates rhythmic periodicity; High (>0.70) indicates random/voluntary noise

Physiological Interpretation & Explainability

Dominant frequency (4.7 Hz) falls precisely within the 4.0 - 6.0 Hz cardinal resting tremor band associated with Parkinsonian motor signs. High spectral energy concentration (75.1%) in the 4-6 Hz band confirms a focused periodic oscillatory pattern, distinguishing it from broadband voluntary limb motion. Spectral entropy (0.48) reflects broad-spectrum kinetic noise or non-rhythmic movement. Dynamic amplitude RMS (0.451 g) and jerk (17.3 g/s) signify substantial biomechanical displacement and rapid oscillatory acceleration.

NOTICE & MEDICAL DISCLAIMER: NeuroTrack is an experimental screening and longitudinal monitoring aid, NOT a diagnostic or medical treatment device. The metrics, classifications, and severity scores provided herein reflect biomechanical time-series observations and must be interpreted solely by a qualified neurologist or physician in clinical context. Never initiate, alter, or discontinue any medication based on this report.